Toy-car investigation: collecting data and graphing

STUDENT INFORMATION AND EXAMPLES

Investigation question

How does the position of a toy car change with elapsed time when it rolls from rest down a shallow, fixed incline?

We will use the same car and the same inclined plane for five trials, measuring elapsed time at marked positions.

Apparatus and procedure

Equipment: toy car, shallow incline with paper scale at 5 cm intervals, ruler, tape, stopwatch, graph paper and pencil.

1. Choose the release mark as x = 0.00 m. Confirm the same ramp set-up and car for all five trials.
2. Release the car from rest without a push. Start timing on release, and record the elapsed time when its front passes each mark.
3. Repeat the whole journey five times. Keep a separate column for each trial. Do not overwrite an unusual reading.
4. For each position, calculate the mean of the five time readings.
5. On A4 graph paper, plot position x (m) vertically against mean elapsed time t (s) horizontally. Label units and draw a smooth best-fit curve.

Gradients of a curved position–time graph

The slope of an x–t graph gives velocity. To estimate instantaneous velocity at an early or late time, draw a tangent to the smooth curve at that time.

Pick two widely spaced points on the tangent, not necessarily actual data points. Calculate gradient = $\Delta x/\Delta t$, in m/s. Explain what each sign and magnitude means.

Near the very first and final measurement, estimated slopes may be less reliable; explain how you selected the tangent.

Data-table headings

Position, x (m)	Time taken to reach position x from the origin (s)					
	t_1	t_2	t_3	t_4	t_5	Mean t
0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.05						
0.10						
0.15						
0.20						